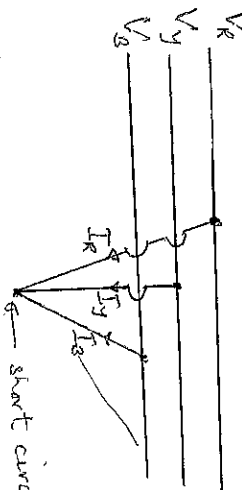


Asymmetrical Fault

Asymmetrical fault occurs when all the three conductors of a 3-phase lines are brought together simultaneously into a short circuit condition as shown in the diagram below.



This type of fault gives rise to symmetrical currents (equal fault currents in all the phases) with 120° displacement. Referring to the diagram fault current I_r , I_y , and I_s will be equal in magnitude.

Unsymmetrical Faults

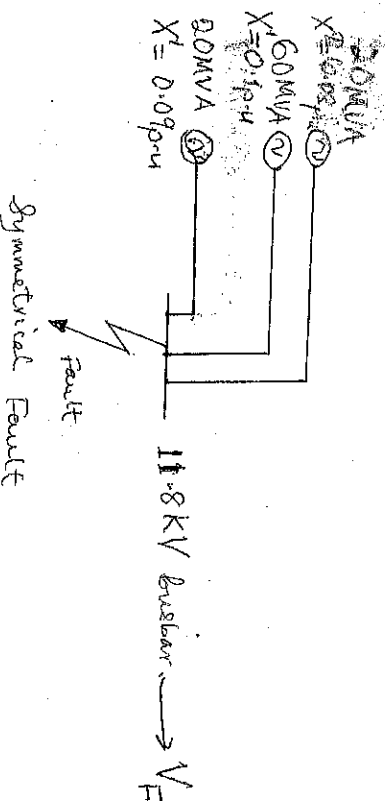
Those faults on the power system which give rise to unsymmetrical fault currents (unequal fault currents in the lines) with unequal phase displacements are known as unsymmetrical faults.

Example:

An 11.8 kV busbar is fed from three synchronous generators having the following ratings and synchronous reactances as follows;

20 MVA, $X = 0.08 \text{ pu}$, 60 MVA, $X = 0.1 \text{ pu}$, and 20 MVA, $X = 0.09 \text{ pu}$.

Calculate the fault MVA level and fault current if a 3-phase symmetrical fault occurs on the busbar.



Solution

Let $(\text{MVA})_{\text{base}} = 60 \text{ MVA}$
 $(\text{kV})_{\text{base}} = 11.8 \text{ kV}$

Transient reactance X'_{eff} for just 20 MVA generator is;

$$X'_{\text{eff}} = \frac{60}{20} \times \frac{0.08}{1} = 0.24 \text{ pu} \quad (10)$$

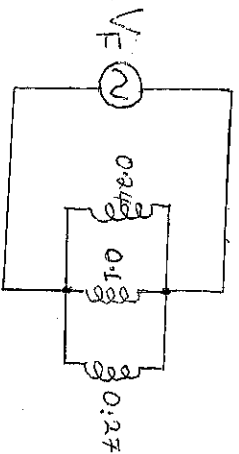
Transient reactance $X'_{(EO)}$ for the second 80 MVA generator is,

$$X'_{(EO)} = \frac{60}{20} \times \frac{0.09}{1} = 0.27 \text{ p.u.}$$

The transient reactance $X'_{(EO)}$ for the 60 MVA generator is,

$$X'_{(EO)} = \frac{60}{60} \times \frac{0.1}{1} = 0.1 \text{ p.u.}$$

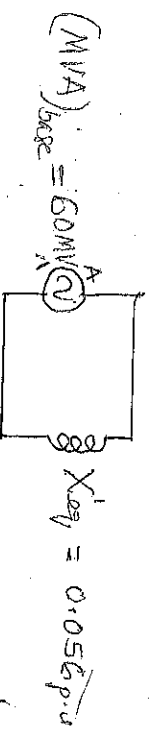
As the generator e.m.f.s are equal, the source may be taken as one and transient reactances connected across one source in parallel as shown,



The equivalent transient reactance is given by X'_{eq} , hence,

$$X'_{eq} = \frac{1}{\frac{1}{0.27} + \frac{1}{0.1} + \frac{1}{0.27}} = 0.056 \text{ p.u.}$$

The final equivalent circuit is shown below;



Hence, the fault MVA level is

$$= \frac{(MVA)_{base}}{X'_{eq}} = \frac{60}{0.056} = 1071 \text{ MVA}$$

The fault current I_F

$$= \frac{\text{Fault MVA level}}{\sqrt{3} \times V_F} = \frac{1071 \times 10^6}{\sqrt{3} \times 11.8 \times 10^3} = \frac{53402 \text{ A}}{11} = 52.4 \text{ kA}$$

Sequence Components

Any unbalanced system of 3-phase currents (or voltages) may be regarded as being composed of three separate sets of balanced vectors, called,

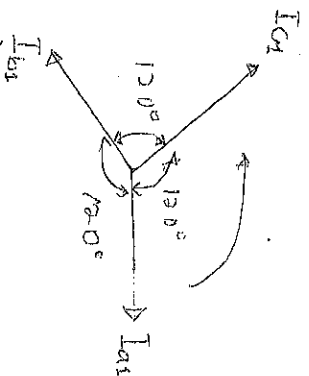
1. positive (or normal) phase sequence components,
2. negative (or opposite) phase sequence components,
3. Zero phase sequence components (three currents equal in magnitude and having zero phase displacement).

Positive Sequence Components

The positive, negative and zero phase sequence components are called the symmetrical components of the original unbalanced system. The subscripts 1, 2, and 0 are generally used to indicate positive, negative and zero phase sequence components respectively.

Negative Sequence Components

It consists of three components of equal magnitude, displacing each other by 120° in phase and having the phase sequence a-b-c.



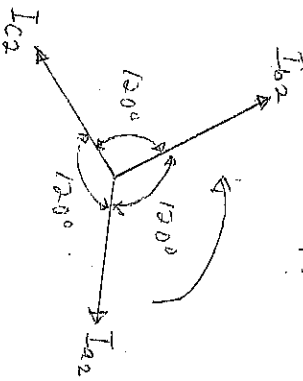
$$I_{a1} = I_{a1} \angle 0^\circ$$

$$I_{b1} = I_{a1} \angle 240^\circ = I_{a1} \angle -120^\circ$$

$$I_{c1} = I_{a1} \angle 120^\circ$$

Zero Sequence Components

It consists of three components of equal magnitude, displacing each other by 120° in phase and having the phase sequence a-b-c.



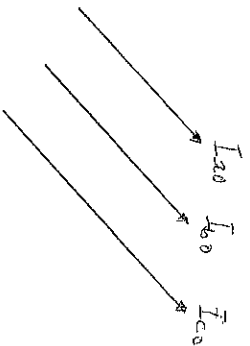
$$I_{a2} = I_{a2} \angle 0^\circ$$

$$I_{b2} = I_{a2} \angle 120^\circ$$

$$I_{c2} = I_{a2} \angle 240^\circ = I_{a2} \angle -120^\circ$$

Zero Sequence Components

It consists of three phases equal in magnitude and with zero phase (0°) displacement from each other.



$$I_{a0} = I_{b0} = I_{c0}$$

Sequence Operator

In unbalanced problem, to find the relationship between phase voltages and phase currents, we use sequence operator a . The operator a is one, which when multiplied to a vector rotates the vector through 120° in the anticlockwise direction. Hence,

$$a = 1 \angle 120^\circ = 1 (\cos 120^\circ + j \sin 120^\circ) = -0.5 + j0.866$$

$$a^2 = 1 \angle 240^\circ = 1 (\cos 240^\circ + j \sin 240^\circ) = -0.5 - j0.866$$

$$a^3 = 1 \angle 360^\circ = 1 (\cos 360^\circ + j \sin 360^\circ) = 1.$$

Therefore

$$1 + a + a^2 = 0, \quad a^3 = 1,$$

$$a^4 = a^3 \cdot a = 1 \cdot a = a$$

Sum of Currents in a Phase

The current in any phase is equal to the vector sum of positive, negative, and zero phase sequence currents as shown.

Red-phase = R

$$\vec{I}_R = \vec{I}_{R1} + \vec{I}_{R2} + \vec{I}_{R0}$$

Yellow-phase = Y

$$\vec{I}_Y = \vec{I}_{Y1} + \vec{I}_{Y2} + \vec{I}_{Y0}$$

Blue-phase = B

$$\vec{I}_B = \vec{I}_{B1} + \vec{I}_{B2} + \vec{I}_{B0}$$

Derivation of Expressions for Symmetrical Components

From the foregoing,

$$\vec{I}_R = \vec{I}_{R1} + \vec{I}_{R2} + \vec{I}_{R0}$$

$$\vec{I}_Y = \vec{I}_{Y1} + \vec{I}_{Y2} + \vec{I}_{Y0} = a^2 \vec{I}_{R1} + a \vec{I}_{R2} + \vec{I}_{R0} \quad (1)$$

$$\vec{I}_B = \vec{I}_{B1} + \vec{I}_{B2} + \vec{I}_{B0} = a \vec{I}_{R1} + a^2 \vec{I}_{R2} + \vec{I}_{R0} \quad (2)$$

$$(3) \text{ Zero Sequence Currents}$$

Adding expressions (1), (2) and (3) together gives,

$$\vec{I}_R + \vec{I}_Y + \vec{I}_B = \vec{I}_{R1} + \vec{I}_{R2} + \vec{I}_{R0} + a^2 \vec{I}_{R1} + a \vec{I}_{R2} + \vec{I}_{R0} + a \vec{I}_{R1} + a^2 \vec{I}_{R2} + \vec{I}_{R0}$$

Collecting like terms together gives,

$$\vec{I}_R + \vec{I}_Y + \vec{I}_B = (\vec{I}_{R1} + a^2 \vec{I}_{R1} + a \vec{I}_{R1}) + (\vec{I}_{R2} + a \vec{I}_{R2} + a^2 \vec{I}_{R2}) + (\vec{I}_{R0} + \vec{I}_{R0} + \vec{I}_{R0})$$

(13)